

Detecting ATM Money Mule Activities via LVLM-Guided Forensic Analysis

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Abstract—We present a Large Vision-Language Model (LVLM)-guided forensic framework for automatic detection of money mule behavior in ATM videos. Money mule networks pose a serious threat to global anti-money laundering (AML) systems, yet existing detection methods are largely transaction-centric and overlook suspicious behaviors observable in video. Recent advances in LVLMs enable extraction of fine-grained behavioral cues from ATM footage, providing complementary evidence beyond financial records. We introduce a Policy-Governed Prompting (PGP) pipeline implementing LVLM inference for money mule detection. The pipeline first uses lightweight body pose recognition modules to efficiently screen long ATM videos, followed by structured LVLM prompts combined with a three-stage reasoning process of Symbolic Grounding, Behavioral Serialization, and Counterfactual Assessment to improve detection accuracy, consistency, and interpretability. We curate a real-world ATM money mule dataset with 30 raw ATM surveillance videos and 47 person-centric interaction clips (30 money mule and 17 normal), organized using a nine-category forensic ontology capturing key behavioral patterns. Experimental results show that our approach reduces LVLM hallucinations and achieves strong performance, with a ROC-AUC of 0.931 and precision of 0.958.

I. INTRODUCTION

Money laundering and the illicit movement of funds through *money mule* networks remain among the most significant threats to the integrity of the global financial system [3]. Money mules are individuals who transfer illegally obtained funds on behalf of criminal organizations act as essential intermediaries during the layering stage of money laundering, obscuring the audit trail between the predicate crime and the final integration of funds [3].

Current detection strategies predominantly rely on transaction-based analysis, leveraging graph neural networks and topology-aware models to identify suspicious fund flows [2], [7]. While effective at mapping digital relationships, these methods encounter a critical physical evidence gap. Specifically, they cannot observe the behavioral anomalies occurring at Automated Teller Machines (ATMs), the primary physical interface where illicit cash enters or exits the banking system. Traditional surveillance systems in these environments often rely on discriminative deep learning models, such as CNN-LSTMs, to detect suspicious motion [1], [6]. However, these models operate as black boxes, providing numerical

anomaly scores without the semantic justification or step-by-step reasoning required for legal forensic auditing and prosecution.

The emergence of Large Vision-Language Models (LVLMs), such as the Gemini-series, offers a promising pathway toward bridging this gap by translating visual signals into logical narratives. Recent advancements in Chain-of-Thought (CoT) reasoning [12] have further enhanced the interpretability of these models. For instance, Structured CoT reasoning has been shown to significantly improve out-of-distribution (OOD) detection in safety-critical road scenes by guiding the model through intermediate rationales [10]. Despite this potential, applying general-purpose LVLMs to financial forensics remains challenging due to the high risk of hallucinations and the lack of alignment with strict investigative protocols. In safety-sensitive domains like anti-money laundering (AML), an automated system must not only detect anomalies but also provide an auditable evidence chain that adheres to expert-defined behavioral ontologies.

To address these challenges, we propose a novel LVLM-guided forensic framework for automated forensic auditing of ATM money mule activities (Figure 1). The framework introduces hierarchical reasoning policy consisting of symbolic grounding, behavioral serialization, and counterfactual risk assessment. It first identifies key environmental entities through symbolic grounding, then maps observed actions into a structured nine-category forensic behavioral ontology, and finally performs conservative, evidence-based classification via counterfactual reasoning. To support practical deployment, we design a lightweight long-video analysis pipeline based on person-centric temporal segmentation, efficiently decomposing continuous surveillance streams into structured interaction episodes, reducing computational overhead while enabling scalable real-world monitoring. Within this framework, a hierarchical reasoning policy constrains LVLM outputs to align with professional forensic standards, reducing hallucinations compared to free-form prompting. In addition, we develop a domain-specific forensic behavioral ontology tailored to ATM money mule detection, capturing subtle behavioral anomalies such as device-switching and surrounding-awareness often overlooked by conventional CNN-based approaches. Experimental results demonstrate that the framework achieves a

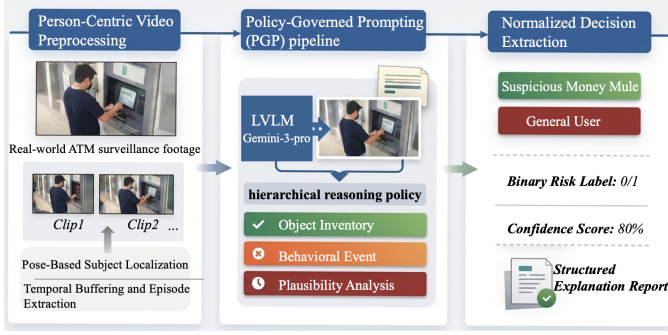


Fig. 1. **LVLM-guided forensic framework for ATM money mule detection.** The overall system operates as a PGP pipeline that transforms raw ATM surveillance footage into auditable case-level decisions. It first performs person-centric temporal segmentation to isolate complete ATM interaction episodes, then applies the PGP module under a hierarchical reasoning policy to generate evidence-grounded behavioral traces, and finally converts the resulting forensic report into standardized decision outputs for evaluation.

ROC-AUC of 0.931 and a precision of 0.958 while producing a human-verifiable evidence chain that satisfies practical financial auditing requirements.

II. RELATED WORK

Financial Crime and Money Mule Detection: The proliferation of money mules has become a major challenge in global financial systems, serving as key intermediaries for laundering illicit funds [3]. Recent research has addressed this problem through network-level and topology-aware modeling. FlowScope detects suspicious transactions using multipartite graph structures [7], while LaundroGraph employs self-supervised graph representation learning for forensic backtracking [2]. Despite these advances in analyzing digital financial traces, a significant evidence gap remains.

Behavior and Anomaly Analysis in ATM Videos: ATM behavioral analysis has progressed from simple motion heuristics to deep-learning-based activity recognition. Traditional approaches rely on discriminative, feature-based methods, while modern video anomaly detection models such as SlowFast [4] and Video Swin Transformers [8] excel at extracting spatiotemporal features. However, these models cannot perform zero-shot detection from natural-language descriptions of criminal behaviors, limiting their usefulness for forensic analysis. Although they can flag anomalies through high-dimensional feature deviation [5], [11], they cannot identify the specific behavioral violations defined by practitioners’ investigative protocols.

LVLMs and Visual Reasoning: Large Vision-Language Models (LVLMs) can connect visual data with textual behavioral descriptions, enabling generalizable and scalable detection of complex illicit activities. Advanced models like Gemini-3-Pro supports long-context video understanding and cross-modal logical inference. Chain-of-Thought (CoT) methods help reduce hallucinations and improve localized reasoning. The foundational CoT framework [12] enables step-by-step reasoning and has been extended to Visual CoT [9] for grounded visual reasoning, as well as Video-of-Thought for

temporal rationale generation. Structured CoT reasoning has further been shown to enhance out-of-distribution detection in safety-critical scenarios by focusing the model on task-relevant regions [10]. While these advances illustrate the potential of inference-driven perception, their application in financial forensics, where reasoning must conform to strict legal ontologies, remains under-explored. Effectively adapting generic LVLMs to detect ATM-related criminal behaviors is still an open challenge, and bridging this gap is crucial for building automated, legally auditable systems capable of identifying subtle behavioral violations.

III. METHODOLOGY

In this section, we present the proposed LVLM-guided forensic framework for ATM money mule detection. The framework is organized as the PGP pipeline that converts raw surveillance footage into auditable case-level decisions. It begins with person-centric ATM interaction segmentation to isolate complete withdrawal episodes from continuous surveillance streams, followed by a Policy-Governed Prompting (PGP) module that performs evidence-first multimodal reasoning under a hierarchical three-stage policy, and ends with a normalized decision extraction layer for quantitative evaluation. Together, these components enable the system to preserve temporal context, generate structured forensic evidence, and produce standardized outputs suitable for both human inspection and statistical assessment.

The proposed pipeline consists of three integrated stages, as illustrated in Fig. 1:

(I) Person-Centric Video Preprocessing and Temporal Segmentation: Raw surveillance streams are segmented into person-centric clips using pose-based detection and temporal buffering. Each clip captures a single, continuous ATM interaction episode, which serves as the fundamental unit for subsequent analysis.

(II) CoT-Governed Evidential Articulation: Each interaction episode is examined following a strict evidential protocol. Guided by carefully designed prompts and chain-of-thought (CoT) reasoning, Gemini-3-Pro-preview is used as the LVLM backbone to enumerate observable objects and record timestamped behavioral events based solely on visual evidence. Importantly, diagnostic judgment is deferred until the evidentiary record is complete.

(III) Normalized Decision Extraction: The final output is encoded in a standardized, machine-readable format, including a binary classification label and a confidence score. By separating evidential reasoning from decision extraction, this design ensures auditability, reproducibility, and consistent evaluation across different prompting regimes.

A. Person-Centric ATM Interaction Segmentation

The raw ATM surveillance stream V often contains environmental noise and irrelevant background activity that can distract a LVLM analysis. To improve both computational efficiency and reasoning precision, we apply *spatio-temporal*

episode localization to isolate high-fidelity interaction segments.

We use YOLOv8-pose as the core detection engine to identify and track human subjects. Unlike standard bounding-box detectors, this pose-based approach provides body keypoints, enabling the system to distinguish active ATM users from background pedestrians. To focus on the primary subject, we apply a *geometric significance* filter:

$$\frac{\text{Area}(B_t)}{\text{Area}(I_{\text{frame}})} \geq \tau_{\text{area}}, \quad (1)$$

where B_t denote the detected person bounding box at frame t , and I_{frame} the total frame area. Setting the empirical threshold $\tau_{\text{area}} = 0.5$ filters out distant subjects and concentrate analysis on individuals near the ATM.

ATM interactions are often subject to transient occlusions or temporary subject departures. To address this, we introduce a *temporal patience* mechanism that maintains continuity despite short-term detection gaps. Starting from the first valid detection, the system tracks each subject as part of an interaction episode s_i and continues the episode as long as the subject reappears within a predefined temporal window, thereby preserving a single, coherent interaction sequence.

To handle intermittent detection failures, such as severe motion blur or pose shifts, we apply a buffer threshold of $\Delta t = 150$ frames. If the subject is lost for fewer than Δt frames, the episode continues; otherwise, it terminates and is exported as a standalone analysis unit. This ensures that each extracted clip $s \in \mathcal{S}$ represents a single, complete ATM transaction episode, providing the temporal context needed for the subsequent reasoning.

B. Policy-Governed Prompting for Evidential Reasoning

The core of our framework is the Policy-Governed Prompting (PGP) pipeline, which defines a hierarchical reasoning policy to guide the LVLm through structured forensic analysis. Unlike traditional black-box classification, PGP leverages Chain-of-Thought (CoT) prompting to enforce a perception-before-judgment workflow, ensuring that each high-risk decision is supported by a human-verifiable, timestamped evidence chain. The CoT prompt is organized into three cognitive stages to reduce hallucinations and prevent premature conclusions:

Stage I: Symbolic Grounding (Object Inventory): The VLM first performs a frame-level inventory of forensic primitives. Key objects, such as *mobile phones*, *wallets*, *NTD cash*, and distinguishing markers like *tattoos*, are identified. This stage maps visual observations to semantic tokens without inferring behavioral intent.

Stage II: Behavioral Serialization: The model then associates identified primitives with actions over time. Each event is recorded with start and end timestamps, along with a concise natural language description (15–30 words), forming a structured sequence of observed behaviors.

Stage III: Conservative Risk Assessment: Diagnostic judgment is deferred until the evidentiary record is complete. The

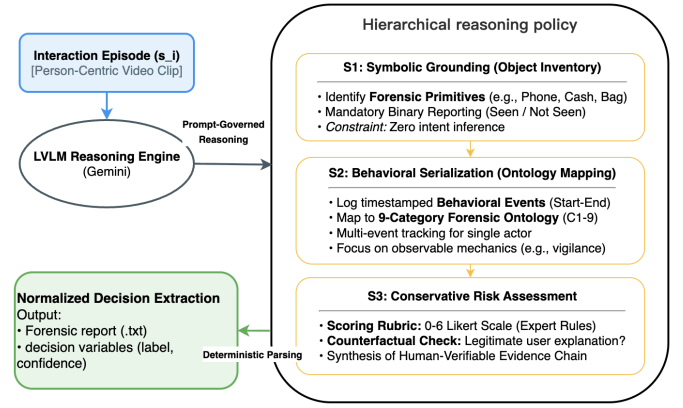


Fig. 2. Overview of the proposed PGP module within the LVLm-guided forensic framework. For each person-centric ATM interaction episode, the module executes a three-stage hierarchical reasoning policy consisting of symbolic grounding, behavioral serialization, and conservative risk assessment.

TABLE I
CATEGORIES, AND KEY VISUAL INDICATORS USED FOR BEHAVIOR-BASED MONEY MULE SCREENING.

Category	Key Visual Indicators
C1	Young appearance and age with high-value cash (> 10 000 TWD).
C2	Stowing large cash volumes immediately without counting or verification.
C3	Use of pre-prepared backpacks or shopping bags instead of wallets.
C4	Explicit visual searching for or direct staring at CCTV camera positions.
C5	Repeated, vigilant scanning (left, right, behind) of the surroundings.
C6	Presence of distinct tattoos on exposed limbs concurrent with other risks.
C7	Continuous mobile device interaction while operating the ATM interface.
C8	Repeated card insertions and PIN entries within a single episode.
C9	Synchronized cash stowage and high-frequency peripheral scanning.

model performs a counterfactual check, assessing whether the observed behaviors can be reasonably explained by legitimate user activity before assigning a suspicious label.

C. Normalized Decision Extraction

To convert descriptive forensic narratives into standardized outputs for quantitative evaluation, we introduce a *Normalized Decision Extraction* layer. For each interaction episode, the preceding PGP stage generates a structured forensic report r , which is preserved as a standalone text artifact and serves as the primary decision-bearing output of the framework. Rather than containing only a final verdict, r records the full analytical result of the case, including observed objects, detected behavioral events, criterion-level risk assessments, final categorical judgment, confidence statement, and a narrative case summary. In this sense, the report functions as the complete evidentiary record for qualitative inspection, case review, and traceable auditing. For quantitative analysis, the report is subsequently normalized into two standardized decision variables: a binary label $\hat{y}$ and a confidence score c . This design explicitly separates evidential reasoning from decision normalization while preserving the original case-level report. As a result, the framework supports both human-readable forensic inspection and large-scale quantitative evaluation.

A discrete classification where 1 denotes suspicious money mule and 0 denotes normal user. This label is derived from

the synthesized risk assessment in Stage III of the reasoning process.

IV. EXPERIMENTAL RESULTS

In this section, we evaluate the proposed PGP pipeline for ATM money mule detection. We first describe the dataset construction process, ground-truth definition, and evaluation setting, and then present quantitative and qualitative comparisons against a simplified prompting baseline. The results show that the proposed method improves evidential consistency, forensic auditability, and class separability, while achieving a more precision-oriented decision profile than direct prompting.

A. Experiment Settings

The experimental evaluation is conducted using real-world ATM surveillance footage provided in de-identified form by the Zhongli Precinct, Taoyuan City Police Department. The dataset originates from confirmed fraud investigations and reflects authentic operational conditions encountered in financial crime enforcement.

From the raw archive, we selected 30 ATM surveillance videos as the basis of our experiments. Each video corresponds to a distinct surveillance session and may contain one confirmed money mule and zero or more ordinary ATM users appearing sequentially within the same recording. No behavioral annotations or role labels were provided to the model during inference.

To ensure consistent units of analysis, all raw videos were subjected to a person-centric preprocessing and temporal segmentation procedure described in Section IV-B. This process isolates continuous ATM interaction episodes based on pose-driven detection and spatial significance thresholds, thereby eliminating irrelevant background footage and interleaved idle intervals. Following this cleaning process, the final evaluation set consists of 30 money mule interaction clips and 17 normal withdrawal interaction clips. Yielding a total of 47 person-centric ATM interaction episodes. Each episode corresponds to a single individual conducting one or more withdrawal actions within a temporally contiguous interval.

Ground-truth labels were assigned based on post-investigation confirmations provided by law enforcement. An interaction episode is labeled as money mule if the individual was identified as participating in organized fraud activities, and as normal user otherwise. Importantly, these labels are used only for post-hoc evaluation, not during prompting or model inference, preserving a strict separation between evidence articulation and performance assessment.

This dataset composition reflects a realistic class imbalance and environmental variability, including differences in camera angles, lighting conditions, withdrawal durations, and user behaviors, thereby providing a stringent testbed for evaluating the robustness of prompt-governed multimodal reasoning under unconstrained surveillance settings.



Fig. 3. Overview of the curated evaluation set after person-centric ATM interaction segmentation, showing the composition of money mule and normal withdrawal episodes used in the experiments.

TABLE II
PERFORMANCE COMPARISON ON THE MONEY-MULE CLASSIFICATION TASK.

Model	N	Accuracy	Precision	Recall	F1	ROC-AUC
PGP Module	47	0.830	0.958	0.767	0.852	0.931
Direct-Prompting Baseline	47	0.723	0.730	0.900	0.806	0.692

B. Quantitative Comparison

The proposed framework was evaluated against a simplified prompting baseline (Direct-Prompting Baseline) that removes the full hierarchical reasoning constraints described in Section III-C. To quantify the effectiveness of the prompt-governed reasoning strategy, we assessed classification performance using standard metrics, including Accuracy, Precision, Recall, F1-score, and ROC-AUC. All evaluations were conducted on the curated dataset of 47 ATM interaction episodes.

Table II presents the quantitative results. The baseline configuration (Direct-Prompting Baseline) achieves an Accuracy of 0.723 and a Recall of 0.900, indicating that it is relatively sensitive in detecting suspicious cases. However, this comes at the cost of lower Precision (0.730) and a limited ROC-AUC of 0.692, suggesting weaker discriminative capability and a higher tendency toward false positives.

In contrast, the proposed Policy-Governed Prompting (PGP) module demonstrates a moderate but consistent improvement over the baseline in most key metrics. It achieves an Accuracy of 0.830 and a higher Precision of 0.958, indicating more reliable flagged cases. However, Recall decreases to 0.767 and F1-score reaches 0.852, reflecting a precision-oriented trade-off. Most notably, the ROC-AUC reaches 0.931, indicating better separability between money mule and normal user behaviors.

C. Impact of Structured Reasoning

The performance improvements can be directly attributed to the introduction of the hierarchical reasoning policy. By enforcing the PGP pipeline: Symbolic Grounding, Behavioral Serialization, and Counterfactual Assessment. The model is constrained to generate evidence before making diagnostic judgments.

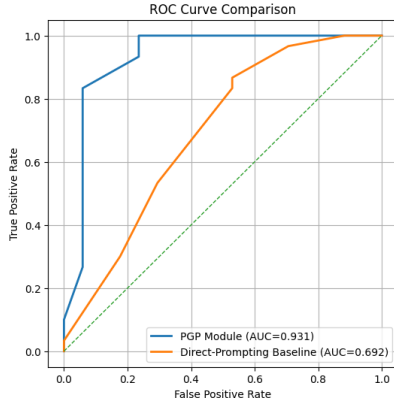


Fig. 4. ROC curve comparison between the proposed PGP module and the Direct-Prompting Baseline.



Fig. 5. Chronological visual trace of one ATM interaction episode. Yellow labels indicate ontology-aligned behavioral codes extracted by the policy-governed reasoning pipeline.

This design reduces premature inference and mitigates hallucination, leading to more conservative and evidence-aligned predictions. Compared to the baseline, which allows more direct mapping from visual input to classification, PGP prioritizes interpretability and evidential consistency, resulting in significantly higher precision and better-calibrated confidence scores.

Figure 5 provides a representative qualitative example. The model first identifies stable contextual entities (ATM interface, hand position, phone presence), then serializes the interaction into timestamped ontology codes (e.g., repeated C5/C9 and C1/C9 transitions) before making a final risk judgment. This evidence-first progression avoids single-frame overreaction and explains why PGP yields fewer false positives than direct prompting.

D. Relative Performance Gains

To further quantify the advantage of the proposed method, we compare PGP against Direct-Prompting Baseline across

key metrics. PGP improves Accuracy by 14.8% (0.830 vs. 0.723) and increases Precision by 31.3% (0.958 vs. 0.730), demonstrating a reduction in false-positive predictions.

Figure 4 further confirms this advantage from a threshold-independent perspective. The ROC curve of PGP remains above that of Direct-Prompting Baseline across most operating regions, indicating stronger separability between money mule and normal behaviors under varying decision thresholds.

In terms of discriminative capability, the ROC-AUC improves by 34.6% (0.931 vs. 0.692), highlighting the effectiveness of structured reasoning in separating suspicious from normal behaviors.

Recall decreases by approximately 14.8% (0.767 vs. 0.900), and F1-score improves by 5.7% (0.852 vs. 0.806). This reflects a conservative decision profile that prioritizes precision over sensitivity in forensic contexts.

E. Forensic Reliability and Interpretability

Beyond numerical performance, a key advantage of the proposed framework lies in its forensic auditability. Unlike the baseline model, which produces direct classification outputs with limited justification, PGP generates structured reasoning traces aligned with the predefined behavioral ontology.

Each prediction is supported by explicit object identification (e.g., cash and mobile device), timestamped behavioral events, and a counterfactual risk assessment grounded in observable evidence.

This evidence-first design allows human investigators to verify model outputs and trace decisions back to concrete visual observations. As a result, the system satisfies not only predictive accuracy requirements but also the interpretability and accountability constraints necessary for real-world AML deployment.

V. CONCLUSION

This study explored the integration of Large Vision-Language Models (LVLMs) with structured visual reasoning to address money mule detection. By implementing the LVLM-guided forensic framework, we demonstrated that an inference-driven approach can effectively bridge the divide between digital transaction logs and observable criminal behavior. Our experimental results, yielding a ROC-AUC of 0.931 and a precision of 0.958, confirm that constraining VLMs with a forensic behavioral ontology significantly suppresses hallucinations and provides the auditable evidence chains required for high-stakes financial auditing.

Despite these advancements, several limitations remain that warrant further investigation. First, the current framework relies on cloud-based LVLM services, which introduces recurring API costs and potential latency issues in real-time surveillance. Second, dependency on stable high-bandwidth communication infrastructure poses a challenge for deploying this system in remote or bandwidth-constrained ATM locations. These factors currently limit the scalability of the system for instantaneous, edge-based processing.

Future research will focus on transitioning toward more autonomous and cost-effective deployment models. A promising direction is the integration of Segment Anything Model 3 (SAM 3) to enhance object-level localization and behavioral tracking. Agentic AI, integration to AIoT devices...

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APPENDIX A EXPERIMENTAL PROMPT

This appendix provides the full prompt used in our experiments. The prompt was given to the LVLM-guided forensic framework to analyze ATM withdrawal videos under the proposed PGP pipeline.

Role and Objective. “You are a forensic investigator specializing in the analysis of Automated Teller Machine (ATM) withdrawal behavior. Your task is to analyze the ATM surveillance videos I provide and determine whether the individual withdrawing cash exhibits a high likelihood of being a fraud-related money mule. You must clearly explain the basis for your judgment. Your primary responsibility is to objectively document observable visual evidence, without speculation or describing non-existent behaviors. Only after completing evidential articulation may you proceed to risk assessment. Do not assume the individual is a criminal; all conclusions must be grounded in verifiable visual evidence.

The videos I provide are ATM surveillance footage. Each video contains a single subject, and may include multiple withdrawal actions by the same individual. No prior annotations or role labels (normal user vs. money mule) are provided. You must independently detect suspicious behavioral patterns and perform assessment based solely on the video.

You must strictly follow the three-stage forensic reasoning protocol. Do not skip stages or mix reasoning steps.”

Stage I: Symbolic Grounding (Object Inventory).

For each video, perform a frame-by-frame objective inventory of observable entities. Identify the following objects: The object inventory includes: mobile phone, wallet, New Taiwan Dollar (NTD) cash, tattoos, backpack, and crossbody bag.

Only record objects when visual evidence is sufficient. Otherwise, explicitly state: “object cannot be confirmed.” Record results in a binary table, where 1 denotes identified and 0 denotes not identified. This stage maps visual observations to forensic primitives without inferring behavioral intent.

Stage II: Behavioral Serialization (Pure Visual Event Logging).

Only describe observable actions. No inference of intention, purpose, or motivation is allowed.

I. Event Detection under Forensic Behavioral Ontology.

For each video, detect and annotate all events related to potential money mule behaviors based on a predefined nine-category forensic behavioral ontology. Each event must include a start time, an end time, and a 15–30 word natural-language description. If an event does not occur, mark it as “n/a.”

Categories C1–C9. C1 records youthful appearance coupled with high-value withdrawals, with explicit notation when age cannot be determined; C2 captures immediate cash stowage without visible verification; C3 covers direct placement of cash into large bags rather than standard wallets; C4 records explicit camera awareness; C5 captures repeated environmental scanning; C6 records tattoos only when co-occurring with other suspicious behaviors; C7 captures mobile-device dependence during ATM operation; C8 records repeated withdrawals within one episode; and C9 captures synchronized cash stowage and vigilance. For all categories, the model must provide timestamps and concise natural-language descriptions, and mark absent events as “n/a.”

Requirement: Evidential Integrity. The model must complete pure factual description first and avoid speculative wording. For example, it may write “the individual carries a black backpack” or “the individual places cash into the backpack,” but it must not write inferred intentions.

Stage III: Conservative Risk Assessment .

Before assigning risk, the model must perform a counterfactual plausibility analysis by asking: “Can this behavior be reasonably explained by a normal user?” If a reasonable explanation exists, the model should prioritize a benign interpretation and reduce the assessed risk. This design aligns with the paper’s counterfactual assessment stage.

II. Money Mule Behavioral Risk Scoring Matrix.

Construct a behavior-based scoring table aligned with the forensic ontology for C1–C9, and overall behavioral deviation. Each item must include a risk score on a 0–6 scale and supporting events with timestamps, or the phrase “not observable or not applicable.” The risk scale is defined as follows: 0 = no suspicion, 1 = slight, 2 = preliminary, 3 = moderate, 4 = high, and 6 = extremely high.

III. Output Structure.

The output must contain the following four parts:

- 1) **Behavioral Event List:** Event ID, category (C1–C9), time range, and description.
- 2) **Risk Scoring Table:** Score and supporting evidence, or “not observable.”
- 3) **Final Classification and Confidence:** Choose either “Suspected money mule” or “Normal user or no clear suspicion.” Confidence is computed from a base of 50%, with +10% per event scored at least 4, +5% for overlapping events, -15% for poor video quality, -10% for limited angle, -10% for short duration, and -20% for contradictory evidence, capped at 95% and floored at 20%.
- 4) **Summary (150–200 words):** Include a behavioral overview, key suspicious patterns, justification of classification, and remaining uncertainties.

Use clear, structured language and avoid unexplained technical abbreviations.